

Algebra 1
Unit 8
Lesson 9 Practice

Name _____
Date _____
Period _____

Florida Real Estate Team just bought a 100 unit apartment building and plans to rent all of the apartments out. The function $P(u) = -10u^2 + 1760u - 50,000$ represents the profit earned, P , depending on the amount of apartment units, u , they rent.

1. Rewrite the function in vertex form.

Complete the statements.

2. The

vertex

 form of the function could be used to determine the interval on which the real estate company would experience a profit.

3. The

vertex

 form of the function can be used to determine when the real estate company experiences the greatest profit.

4. The company is predicted to experience the greatest profit when _____ units are rented, resulting in a profit of \$_____.

Adam is coding a new game where darts are launched at a moving target. He enters the function $H(t) = -0.5t^2 + 6t - 10$ to model the height, H , of the darts in relationship to the ground after t seconds.

5. Rewrite the function in vertex form.

6. Rewrite the function in factored form.

Complete the statements.

7. The

☐ vertex

☐ factored

☐ standard

 form of the function can be used to determine how long the dart is above the ground. The dart is in the air for _____.

8. The dart was launched from a point _____ units

☐ above

☐ below

 the ground and went

☐ above

☐ below

 ground after _____ seconds.

9. The

☐ vertex

☐ factored

☐ standard

 form of the function can be used to determine when the dart reaches its highest point.

10. The dart reaches its highest point of _____ units _____ seconds after it is launched.